Measuring Cross-Country Differences in Misallocation

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This paper is a revised version of *Measuring Cross-Country Differences in Misallocation (CES 16-50) from November, 2016. A copy of the original paper is available upon request.*

CES 16-50R September, 2020

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Abstract

We describe differences between the commonly used version of the U.S. Census of Manufactures available at the FSRDCs and what establishments themselves report. The originally reported data has substantially more dispersion in measured establishment productivity. Measured allocative efficiency is substantially higher in the cleaned data than the raw data: 4x higher in 2002, 20x in 2007, and 80x in 2012. Many of the important editing strategies at the Census, including industry analysts' manual edits and edits using tax records, are infeasible in non-U.S. datasets. We describe a new Bayesian approach for editing and imputation that can be used across contexts.

[*] Some of the research in this paper was conducted while the second author was an employee of Census Bureau. The views expressed in this paper are those of the authors and not the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. The paper includes output approved for release under Disclosure Review Board release numbers DRB-B0115-CDAR-20181016 and CBDRB-FY20-CES007-002 and DRB Bypass number CBDRB-FY19-CMS-7909. We are grateful to many colleagues and seminar audiences for their feedback. Thanks to Mitsu Nishida and Amil Petrin, with whom we started this project, for Hang Kim for his helpful comments, to Allan Collard-Wexler for a thougtful discussion, and to Lee Tanenbaum and Weitao Lin for their expert research assistance.

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I Introduction

Over the past twenty years, many economists (including us) have written papers and lecture notes highlighting that within-industry misallocation of factors can help explain cross-country differences in productivity (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). The source of this belief is a robust stylized fact that developing countries like China and India have more measured within-sector dispersion in establishment behavior than wealthier countries like the US. This paper describes some challenges with the measurement of this stylized fact.

The confidence we have in our claims about dispersion in establishment behavior - either the "true" values for a particular country, or of cross-country differences - depends on how worried we are about measurement error. In principle, measurement error has an ambiguous effect on measured dispersion, since non-classical noise can either push establishment' reported values towards or away from what is typical in their sector. In this paper, we discuss two potential sources of measurement error: establishments potentially misreporting their own characteristics, and subsequent data processing potentially introducing new errors.

Most statistics agencies initially ask firms to verify (or send in) information, but the subsequent steps vary across surveys. Many statistical agencies in developed countries, including the U.S. Census Bureau, both edit and impute responses.[1] Over half of establishments have at least one characteristic which is affected by the cleaning process[2]. The exact procedures vary across industries and time (White et al., 2018), but broadly take two forms. First, the Census Bureau *edits* some outliers. If a reported variable fails one

[1] Less developed countries are less likely to directly edit the raw data. For instance, for India and China we have confirmed that there was no editing or imputation of the data (both in the data documentation, and in email communications with the relevant national statistics agencies). There is some imputation of prices in the Indian data post-2006, but we do not use the price information in our results.

[2] Throughout the paper, we set aside edits to capital given well-known conceptual difficulties with measuring it correctly (Hicks, 1981; Hulten, 1991; Pritchett, 2000; Collard-Wexler and De Loecker, 2016).

or more edit rules, then it may be temporarily replaced with a missing value. Second, the Census Bureau *imputes* missing information, using other information reported by the plant (both in that year and in previous years) and other plants in the same industry.[3] For 2002, 2007, and 2012 we have access to the original and cleaned values reported by firms for plants in the Census of Manufactures,[4] as well as the relevant edit flags.

We have three mains goals in this paper. Our first goal is to demonstrate the extent of data cleaning efforts undertaken by the U.S. Census Bureau. Both responses to obvious reporting errors (such as when establishments report distinct values for the same outcome), as well as more subjective edits (such as manual edits by industry experts) have large effects on measured dispersion. We compare the original responses to the cleaned responses in a variety of ways, focusing on two measures of productivity: revenue productivity (TFPR) and quantity productivity (TFPQ).[5] For most establishments, at least one of the variables needed to calculate TFP is imputed (White et al., 2018). First, we show visually that cleaning has large effects on the distributions, dramatically lowering the mass in the tails while also shifting the distributions up for TFPR.

Most quantitative measures of misallocation use statistics that are broadly similar to the standard deviation in revenue productivity (TFPR) (Asker et al., 2019). These types of measures are sensitive to tails, and indeed we find that the standard deviation of TFPR falls by half in the cleaned data. However, the data processing undertaken by the U.S. Census Bureau does not only affect the tails: it lowers the measured interquartile range of revenue productivity by almost as much. Both the effect of data processing and measured

[3] Editing a variable normally involves deleting the original response, and then treating it as if it had never been reported.

[4] It is worth noting that when Hsieh and Klenow (2009) was published, neither imputation flags nor the original firm responses were available for the Census of Manufacturers. Interested researchers now can get access in RDCs.

[5] There are well-known difficulties in measuring TFPQ and TFPR, both because estimating production functions is difficult even with data on quantities, and because in many datasets the only measure of output is revenues. We use simple methods to estimate TFPQ and TFPR. In ongoing work we are collecting longer panel data, which will allow us to say more about measuring production function elasticities.

dispersion (in the raw & cleaned data) are increasing over time.

The data cleaning matters most for smaller and younger plants: the average absolute difference between captured and cleaned TFPR for the largest (or oldest) plants is around 2/3 of that for the smallest (or youngest) plants, with a fairly monotonic relationship in between. There is a similar (though smaller) relationship for TFPQ. Nevertheless, there is still a large difference between captured and cleaned productivity even for the largest and oldest plants.

In many models (e.g. Hsieh and Klenow 2009) the covariance of TFPR and TFPQ is important for measuring misallocation, with the logic that distortions matter more for aggregate activity if they affect establishments who would be large in the counterfactual absence of distortions. In order to operationalize this assumption, we use the same type of model as in Bils et al. (2017) and Hsieh and Klenow (2009), where consumers have CES demand, and establishments have Cobb-Douglas production functions and face idiosyncratic distortions on their inputs. This generates relatively intuitive expressions for the aggregate productivity losses because of those frictions. If instead of using the U.S. Census Bureau's cleaned data we simply trim the 1% extremes in the data originally reported by each establishment, we find that moving to the new measured U.S. efficiency would *decrease* measured (Hsieh and Klenow, 2009) TFP by a factor of 8 for the Indian formal sector. We do not take this result literally - we do not think that raw data gives compelling evidence that the U.S. manufacturing sector is characterized by more misallocation than India. Instead, we consider our results a "smoking gun" that measurement (and data processing in particular) is deeply important to the study of misallocation.

Given that the overall cleaning effort has a large effect on measured misallocation, we turn to describing the specific process of cleaning the data. After qualitatively describing the different types of edits, we assess their effects by measuring how much each edit type affects measured dispersion. We do so by describing their Shapley (1953) value;

essentially the share of the total change that can be attributed to each type of edit.

The most important edits are "logical" edits and analyst corrections. Logical edits are possible because the Census implicitly asks for the same information in multiple ways, for instance by asking for the plant's total value of shipments as well as the total value of shipments for each product. If the two values for the same outcome diverge, the Census may edit the reported total with the sum of the disaggregated components. Analyst corrections rely on the expertise of full-time industry specialists employed by the U.S. Census Bureau.

In many international datasets with microdata on establishments, many types of processing done by the U.S. Census Bureau are infeasible.[6] For instance, logical edits can only be implemented when there are redundant questions. Given the importance of these edits, it is difficult to interpret measured cross-country differences that use different processing methods. Given the extent of measurement error, it is also difficult to trust differences in unprocessed data. Since we are not nihilists, we describe the effect of methods which commonly clean data across contexts. Trimming is a popular data cleaning strategy which can easily be implemented. However, it is a blunt instrument, and varies paper to paper: for instance while Hsieh and Klenow (2009) trim static measured distortions, other papers using the exact same data have trimmed other characteristics such as growth rates (Allcott et al., 2016; Martin et al., 2017). We describe and implement a more reproducible approach: a theoretically-motivated data cleaning exercise which could then be used across establishment-level datasets without further need for data processing (Kim et al., 2015).

Unlike trimming, which drops outliers and leaves missing values blank, the Kim et al. (2015) method simultaneously edits and imputes the data. First, we look at the ratios of

[6] There are also differences in enumeration, for instance in the United States the Census is filled out using a web portal whereas few Indian manufacturing plants report having computers.

reported variables and flag the outliers of the ratios–this is a standard first step in the literature (Fellegi and Holt, 1976; Thompson et al., 2004). This step unfortunately remains a little ad-hoc across countries. We use the actual bounds used by the U.S. Census Bureau, and try to approximate their equivalents in the Indian data.[7] We then impute entries in order for the cleaned data to pass the edit checks. Unlike the imputation methods the Census Bureau uses most frequently in the Census of Manufactures, the Kim et al. (2015) method tries to preserve the joint distribution of the covariates, using a Bayesian approach.[8] Given the ratio outlier flags, we favor making edits that are likely given our model for misreporting, and similarly impute values that are likely given the underlying model for the data. The imputation step works for missing values as well as those which are flagged as outliers.

Economists have a long tradition of studying (mis)measurement. Our paper is firmly in the spirit of (Romer, 1986a,b, 1989), who study how differences in data quality matter for understanding the historical incidence of business cycles and unemployment.[9] We also complement current efforts to link the Current Population Survey to administrative records in order to, for instance, more reliably describe poverty in the US (Meyer and Mittag, 2019; Medalia et al., 2019). In the misallocation (Banerjee and Duflo, 2005; Restuccia and Rogerson, 2008; Hopenhayn, 2014) literature in particular, Bils et al. (2017); Gollin and Udry (2019) and Esfahani (2019) all study the role of measurement. Their approaches use economic theory to distinguish between measurement and "true" misallocation, for instance by arguing that farmers are unlikely to misallocate resources across their own

[7] There are many reasons why it is impossible and unhelpful to use the same bounds in the US and India. One reason is that the US bounds are industry specific, and the industry classifications are different across countries. Another is that some of the bounds are nominal (e.g. wage bill per worker), for which the underlying units are different across countries.

[8] The Census methods of imputation of missing data also lower measured misallocation relative to more flexible approaches (White et al., 2018).

[9] Similar issues have recently been discussed in the asset pricing literature, since measuring the correlation of consumption with stocks is difficult (Savov, 2011; Kroencke, 2017).

plots. One value of our approach is that we leverage expert editing done by those with their own goals for creating reliable data. Furthermore, researchers can use our proposed data cleaning procedure regardless of their question, and even in settings with less rich data collection and cleaning efforts than the US.

In the next section, we recap the theory of distortions underlying much of our analysis. Section 3 discusses data collection and cleaning procedures for manufacturing data in the United States. Section 4 describes the Kim et al. (2015) method for cleaning plant-level data. In Section 5, we compare commonly-cleaned data for the U.S. and India. Section 6 concludes.

II A Theory of Misallocation

In order to keep our results parsimonious, we focus on describing how TFPR and TFPQ are affected by data processing. Measuring the former requires estimating production functions, which we do using cost shares (recognizing that even if establishments actually had constant returns to scale Cobb-Douglas production functions, cost-shares are only an unbiased measure of production function elasticities in the absence of frictions). Measuring the latter requires a model of demand. We follow Hsieh and Klenow (2009) in imposing CES demand, although like Wang (2019) and Bils et al. (2017) we additionally allow for frictions on intermediate goods.[10]

First, we start from the firm's problem, showing how firm behavior is affected by idiosyncratic distortions on capital and output. In the model, variation in those distortions is captured by variation in firm-level revenue productivity. We then turn to the (static) equilibrium, and derive how aggregate productivity would be affected in a counterfactual where the variation in revenue productivity is removed, which is our measure of misallocation.

[10] For our descriptions of the theory, we refer to firms. In the data, we estimate frictions and production functions at the establishment (i.e., plant) level.

II.A Firm-level Distortions

Final good production Q is a Cobb-Douglas aggregate over sectoral gross output Q_s,

$$Q = \prod Q_s^{\theta_s}.$$

Normalizing the price of the final good, P, to 1, expenditure for each sector is a fixed proportion

$$P_s Q_s = \theta_s Q$$

where P_s is the price index for sector s. The final good can either be consumed or used in a roundabout fashion as an intermediate input (Krugman and Venables, 1995): $Q = C + M$.

All firms use the same intermediate input, denoted M_{si}, so $M = \sum_{s=1}^{S} \sum_{i=1}^{N_s} M_{si}$. Within each sector, output takes a CES form over output of each variety Q_{si}:

$$Q_s = \left(\sum_{i=1}^{N_s} Q_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

and each firm produces gross output using capital, labor, and the intermediate input with Cobb-Douglas production-function elasticities which vary across sectors:

$$Q_{si} = A_{si}(K_{si}^{\alpha_s} L_{si}^{1-\alpha_s})^{\gamma_s} M_{si}^{1-\gamma_s}.$$

The wage and rental rate are constant in the economy, but firms face idiosyncratic distortions on labor, capital, and intermediate inputs. As a result, each firm's profits are:

$$\pi_{si} = P_{si} Q_{si} - \left(1 + \tau_{L_{si}}\right) w L_{si} - \left(1 + \tau_{K_{si}}\right) R K_{si} - (1 + \tau_{M_{si}}) P M_{si}.$$

II.A.1 Optimization Behavior

Firms are monopolistically competitive and face downward-sloping demand curves given by $Q_{si} = Q_s \left(\frac{P_{si}}{P_s}\right)^{-\sigma}$. Profit maximization implies that the firm's output price is a fixed markup over its marginal cost and so there is complete pass-through of improvements in TFPQ (A_{si}). As a result, revenue productivity, $TFPR_{si} = \frac{P_{si}Q_{si}}{\left(K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}\right)^{\gamma_s} M_{si}^{1-\gamma_s}}$, only varies due to the distortions. In particular,

$$TFPR_{si} = \frac{\sigma}{\sigma - 1} \left[\left(\frac{\left(1 + \tau_{K_{si}}\right) R}{\gamma_s \alpha_s} \right)^{\alpha} \left(\frac{\left(1 + \tau_{L_{si}}\right) w}{\gamma_s \left(1 - \alpha_s\right)} \right)^{1-\alpha} \right]^{\gamma_s} \left[\frac{\left(1 + \tau_{M_{si}}\right)}{\left(1 - \gamma_s\right)} \right]^{(1-\gamma_s)}. \quad (1)$$

Similarly, TFPQ can be inferred by taking advantage of the fact that the markup is known:

$$TFPQ_{si} \equiv A_{si} \propto \frac{\left(P_{si}Q_{si}\right)^{\frac{\sigma}{\sigma-1}}}{\left(K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}\right)^{\gamma_s} M^{1-\gamma_s}}$$

While we focus most of our attention on TFPQ and TFPR measured using gross-output production functions, Hsieh and Klenow (2009) using a value added specification, and we show how their results change as a function of the data cleaning as well. The value added approach (where we follow a value-added approach, but firms actually make choices as a function of gross output) gives

$$TFPR_{si}^{VA} = \frac{\sigma^{VA}}{\sigma^{VA} - 1} \left(\frac{\left(1 + \tau_{K_{si}}\right) R}{\gamma_s \alpha_s} \right)^{\alpha} \left(\frac{\left(1 + \tau_{L_{si}}\right) w}{\gamma_s \left(1 - \alpha_s\right)} \right)^{1-\alpha} \left[1 - \frac{\left(1 - \gamma_s\right)}{\left(1 + \tau_{M_{si}}\right)} \right] \quad (2)$$

and

$$TFPQ_{si}^{VA} \propto \frac{\left(P_{si}Q_{si} - M\right)^{\frac{\sigma^{VA}}{\sigma^{VA}-1}}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}}.$$

In the next subsection, we show how variation in TFPR affects aggregate productivity.

II.B Aggregate Distortions

Aggregate productivity in each sector is

$$TFPQ_s = \frac{Q_s}{(K_s^{\alpha_s} L_s^{1-\alpha_s})^{\gamma_s} M_s^{1-\gamma_s}} \equiv \frac{\overline{TFPR}_s}{P_s}, \tag{3}$$

where, given cost-minimization, the price index for sector s is:

$$P_s = \left(\sum_{i=1}^{M} P_{si}^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \left(\sum_{i=1}^{M} \left(\frac{A_{si}}{TFPR_{si}} \right)^{\sigma-1} \right)^{\frac{1}{1-\sigma}}.$$

Plugging back in to Equation 3 gives the core Hsieh and Klenow (2009) expression for productivity

$$TFP_s = \left(\sum_{i=1}^{M} \left(A_{si} \widetilde{TFPR}_{si} \right)^{\sigma-1} \right)^{\frac{1}{\sigma-1}}, \tag{4}$$

where $\widetilde{TFPR}_{si} = \frac{\overline{TFPR}_s}{TFPR_{si}}$. Since we know from Equation 1 that $TFPR_{si}$ would only be different from $\overline{TFPR}_s$ in the presence of distortions, the "efficient" counterfactual TFP is $\overline{A}_s = \left(\sum_{i=1}^{M} A_{si}^{\sigma-1} \right)^{\frac{1}{1-\sigma}}$. Aggregating over all sectors,

$$\widetilde{TFP} = \frac{TFP}{TFP_{(efficient)}} = \prod_{s=1}^{S} \left[\sum_{i=1}^{M_s} \left(\widetilde{TFPQ}_{si} \times \widetilde{TFPR}_{si} \right)^{\sigma-1} \right]^{\frac{\theta_s}{\sigma-1}}, \tag{5}$$

where $\widetilde{TFPQ}_{si} = \frac{A_{si}}{\overline{A}_s}$. The index in Equation 5 ranges from 0 to 1 and can be calculated from observed data. The three main outcomes that we describe in the paper are $\ln(\widetilde{TFPR}_{si})$ and $\ln(\widetilde{TFPQ}_{si})$, and $\widetilde{TFP}$. For notational convenience, although we only present results for log and industry-scaled productivity measures, we describe them as "TFPR" and "TFPQ" in our results.

Instead of measuring how sensitive our three measures of productivity are to differ-

ent underlying assumptions, which has been the primary focus of much of the recent methodological literature on misallocation (Haltiwanger et al., 2018; Asker et al., 2014), we instead calculate them using different cuts of the data, which we describe in the next section.

We also describe which types of establishments experience a larger change in measured characteristics when comparing the captured to the final data. We focus on two measures, the age and the number of employees of the firm. For both, we run a local polynomial regression comparing the difference (either regular difference or the absolute difference) between measured and captured TFPR and TFPQ.

III Data Cleaning in the United States

We primarily use micro-data from the United States, from the 2002, 2007 and 2012 U.S. Censuses of Manufactures. The quinquennial survey covers roughly 300,000 manufacturing plants.[11] Between those years, the Annual Survey of Manufactures collects information on a rotating panel of establishments, with the largest establishments surveyed with certainty, and smaller plants in the frame with some probability (which is increasing with size). The sampling weights are available in the U.S. Censuses of Manufactures,[12] and we show how the measures change when using the sample instead of the Census.

As in most surveys, not all respondents answer all of the questions, and some responses are inconsistent with each other or inconsistent with administrative records data (primarily IRS payroll tax records) from the same firms. The Census Bureau has created imputation and edit rules for this data, the development of which are described in Sigman (1997) and Thompson and Sigman (1999). However, until recently, it was difficult for researchers to identify which, if any, responses for a given plant were imputed.[13] Us-

[11] Information for plants with fewer than 5 employees - roughly one third of the sample - are almost entirely imputed. The standard is to exclude these so-called administrative records plants (Foster et al., 2016), and we follow that standard throughout our analysis.

[12] Sample are selected at 5-year intervals beginning in years ending with 4 or 9.

[13] Item-level edit/impute flags for the 2002 and later censuses became available to researchers a few years

ing the recently available imputation flags, White et al. (2018) shows that over 70% of observations in the Census of Manufactures have imputed data for at least one variable used to compute TFPR, and within-industry TFPR dispersion is significantly smaller in the imputed data than in the non-imputed data. We go beyond the imputation flags and use the newly available actual responses from the establishments themselves (the "raw" data).

We focus on the four variables used by Hsieh and Klenow (2009) to measure plant-level total factor productivity: total value of shipments, total cost of materials and energy, total salaries and wages, and the capital stock. The first three variables are directly measured in the Census of Manufactures, and are present in both the raw data and the final data. For the capital stock, we use the real capital stock variable constructed (from the final data) as part of the BLS-Census Multifactor Productivity project, as described in Cunningham et al. (2018). This variable is not available in raw data, so we use this measure of the capital stock for all of our TFPR and TFPQ calculations.

The raw data differs from the final ("cleaned") data in two respects.[14] First, missing values due to non-response in the reported data are imputed in the cleaned data, using a variety of industry-specific regression-based and other imputation strategies. Second, actual responses which fail edit rules in the reported data are also imputed or changed in some way in the final data. The most important edit rules are balance edit rules (certain variables have to add up) and ratio edit rules (ratios of certain variables must be within certain bounds). Edit-rule-failing responses are replaced using a variety of methods, described in Table 1, and the ones that affect measured misallocation the most are described more fully in Subsection IV.C. We formally describe the logic of data cleaning in Section

ago, and item-level flags for the 1987, 1992, and 1997 censuses became available to researchers in the Federal Statistical Research Data Centers (FSRDCs) in November 2018.

[14] Another convention for naming the data is "captured" data for the reported information, and "completed" for the cleaned data.

V.

There are a few important parameters that are not directly reported in the data: the production function elasticities and the demand elasticity of substitution. We use cost shares for the former, and 4 for the latter (Redding and Weinstein, 2020).[15]

In the next subsections, we compare the clean and raw versions of the data. First we show the differences in the distribution of productivity, then in measures of dispersion, and finally on measures of misallocation.

III.A The distribution of TFPR and TFPQ

We start by describing how data processing affects the measured density of (scaled) TFPR. We do this in the following way. First, we create bins for each 2% of the distribution of TFPR and TFPQ in the final data. Then, in each of these bins, we plot the difference in density between the raw and clean data (so positive values means relatively more density in the bin in the raw data).[16] The differences-in-densities are plotted in Figure 1, with the left graph showing the difference in densities for TFPR, and the right graph for TFPQ.

For TFPR, in all three years there is more mass in the raw data in the tails, which is presumably consistent with most forms of data cleaning that try to address outliers. However, at the lower end this is true only for the lowest (two) bins. All three years exhibit a check-mark shape, where the lower-middle of the distribution is over-represented in the clean data, and the extremes, especially the top of the distribution, are over-represented in the raw data.

The TFPQ difference in density in graphs are essentially flipped: there is more density in the raw data for the lowest values of TFPQ, with no qualitative change over time.

[15] We use an elasticity of substitution of 3 for the Value Added results.

[16] In order to limit the disclosure of the data, we slightly jitter the exact end points of each bin, so readers cannot look with a magnifying class to identify the TFPR of, e.g. the 22nd percentile firm. Similarly, we move the density in the top and bottom bins into their respective neighbors, so we are only plotting 48 bins.

IV Who gets edited?

In this section, we describe which types of firms see large differences between their captured and final data, focusing on age[17] and employment (we use measures of firm characteristics, although we note that most firms are single-establishment firms). In Figure 2, we plot the relationship between the absolute difference in measured TFPR between the captured and final data.[18] Figure 3 plots the absolute differences for TFPQ. The pattern is similar for the three years for which we have data; the absolute difference is consistently falling in both firm age and firm size. The average absolute difference for the largest firms is large - and is often around 50% - but it is larger for the younger and smaller firms.

The absolute gap could be positive even if the cleaning were mean zero. Figure 4 shows that the edited data consistently has smaller measured TFPR (and TFPQ) than the captured data. Again, the size of the average gap is decreasing in firm size and age, although the pattern for firm age is somewhat flatter. Figure 5 shows that for TFPQ, the larger and older firms have larger gaps than their peers (except for 2007, which is fairly flat).

IV.A Dispersion of TFPR and TFPQ

In this section, we quantify the results from the previous subsections. Specifically, we describe the effects of data cleaning on alternative measures of dispersion, specifically the 90/10 ratio, the interquartile range, and the standard deviation.[19] For simplicity, we report values with no trimming. The first column in Table 2 reports the standard deviation. For TFPR, the standard deviation in the cleaned data is around half of that in the original data. Measured dispersion increases over time, although by relatively more in

[17] Firm age comes from the Census Bureau's Longitudinal Business Database. That dataset begins with the year 1976, so the age of firms born before 1976 is censored.

[18] For these graphs, to comply with Census Bureau disclosure avoidance practice, we drop the 1% outliers of TFPQ and TFPR from the graphs.

[19] Recall that we are reporting TFPR and TFPQ measures in logs (and industry-scaled). Thus these 90/10 and 75/25 "ratios" are actually log-differences within each industry, averaged across all industries.

the captured data: in the captured data, the standard deviation increased by over 20% from 2002 to 2012. In the final data, it increased by under 10%. The change from captured to cleaned data is even larger for TFPQ, with the cleaned data having around a third as much dispersion as the captured data. There is a less clear time trend.

We see a similar pattern for the 90/10 ratio and the interquartile range. The captured data has 2-3 times more spread than the final data does, both for TFPR and TFPQ. Even the interquartile range is dramatically affected by the data cleaning undertaken by the U.S. Census: the gross output range in 2012 fell by almost a factor of three. Similar to the standard deviation, there have been a larger increases over time in the captured data than in the final data.

Taken together, these results show that the cleaning undertaken by the U.S. census is more nuanced than trimming tails, since it dramatically effects ratios of interior quantiles.

IV.B Measured Misallocation in the Raw U.S. data

For our final set of results comparing the cleaned and captured data, we use the model described in Section 2. While further removed from the raw data, the advantage of the calculation is that it gets closer to thinking about (measured) welfare costs: in most models, frictions are particularly important if they affect establishments' firm size ranking (Hopenhayn, 2014), which in the undistorted equilibrium is a function of (only) TFPQ.

First, we consider the effects on measured misallocation of replacing cleaned with raw data in the U.S. manufacturing sector in 2002, 2007 and 2012. Table 3 shows the results of calculations across a range of similar datasets.[20] For gross output, in the raw data with no trimming, allocative efficiency is nearly 0 and gets worse (lower) in 2007 and 2012. Trimming the 1% tails raises measured allocative efficiency to 0.109 in 2002, implying that the US manufacturing sector was about 11% as productive as it would be if there were no

[20] For all the results reported in the paper we set the elasticity of substitution $\sigma = 3$ when considering value added production functions, and $\sigma = 4$ for gross output (Hsieh and Klenow, 2009; Bils et al., 2017).

distortions. With this level of trimming, allocative efficiency still falls to 0.012 in 2007 and 0.004 in 2012. We see a similar pattern with 2% trimming except that allocative efficiency levels off in 2012. When we use the Census-cleaned data, we find a similar pattern over time for all 3 levels of trimming – allocative efficiency falls from 2002 to 2007 and levels off or increases slightly from 2007 to 2012. The data cleaning done by the Census Bureau has an enormous effect on measured allocative efficiency. With 1% trimming, measured allocative efficiency ranges from 4 to 87 times higher in the cleaned data versus the raw data.

Using a value added specification, as in Hsieh and Klenow (2009), requires dropping establishments with negative value added (around 2% of the sample in the captured data and 1% in the cleaned data). The potential gains from reallocation are subsequently much smaller than the gross-output counterparts, but the relative effect of cleaning is substantively similar, with an order of magnitude less misallocation in the untrimmed data. Trimming brings the values for the captured and cleaned data closer, but measured allocative efficiency is roughly twice as high in the cleaned data.

The Census of Manufactures is only undertaken quinquenially. In other years the detailed manufacturing data only covers a sample of plants.[21] In Table 4, we report the measured misallocation numbers from the Annual Survey of Manufactures sample within the census year, where each plant is weighted by its ASM sample weight.[22] Both for value added and for gross output production functions, the switch from census to sample has little effect on measured dispersion in the Census-Cleaned data. However, in the raw data there tends to be somewhat less measured misallocation in the sample relative to the census.

[21] In the Annual Survey of Manufactures, plants above a certain size are sampled with certainty every year. Below the size threshold, plants are sampled with probability roughly proportional to size in a 5-year rotating panel.

[22] Due to small differences between samples, we could not disclose the 2012 value added number with 2% trimming.

IV.B.1 Measured Cross-Country Differences in Misallocation

We now turn to discussing cross-country differences in measured misallocation. While we have measured misallocation in the United Sates for a large set of data choices, in order to avoid tedium we only describe cross-country differences in misallocation for two extremes: the average of 2002, 2007, and 2012 for Census-cleaned data, and the corresponding average in the Census raw data.[23] We compare these averages to (Gross Output) estimates from India's Annual Survey of Industries in 2002 and 2007.[24] The results are shown in Table 5. While estimated allocative efficiency in India in 2002 and 2007 is slightly higher than the average for the cleaned U.S. data, it is over 8 times higher than average measured allocative efficiency in raw U.S. data. Taken literally, the latter result would mean that (if there was no measurement error) the Indian manufacturing sector would be only an eighth as productive as it is if it had the same allocative efficiency as the U.S. manufacturing sector.

IV.C The effect of edits on measured misallocation in the raw U.S. data

There are many different data cleaning steps on the path from measured allocative efficiency of 0.004 in the 2012 raw data to 0.349 in the corresponding cleaned data. In order to determine if changes to the raw data are needed, the Census Bureau primarily uses balance edit rules and ratio edit rules. Raw data that fails one or more of the edit rules is replaced using one of the edits/imputations described in Table 1

In this subsection, we characterize the effect of each edit. For eight edit supercategories, we measure misallocation in the U.S. using raw data for everything but those ed-

[23] For all values, when possible, we use the reported values from trimming the 1% extremes for TFPQ and the distortions for capital, labor and intermediate inputs.

[24] The ASI is described in Appendix A.I. Most Indian manufacturing establishments (and employment) are in the informal sector, which are covered in the NSS Schedule 2.2. Most labor for informal establishments is owner labor - with a substantial additional share coming from workers who are neither owners nor hired. As a result, the reported wage bill understates labor inputs. For this particular project, we focus on the formal firms in order to avoid needing to impute. However, underreported owner wages may also be an additional source of measurement error even in the U.S. data (Smith et al., 2019).

its. The eight supercategories are: logical edits (separately for shipments, materials, and payroll), administrative edits, regression imputes, rounding imputes, analyst corrections, and the rest. We describe each supercategory in turn.

Logical edits are done when there are multiple survey questions which ask for the same information. For instance, total value of shipments shows up in three different parts of the survey: (1) there is a question that asks for the total value of the plants shipments; (2) there are many questions about the value of shipments for specific products that a given industry produces (these values can be summed by the U.S. Census Bureau); and (3) there is a question – separate from (1) – that asks the respondent to total the values of the products in (2). If these values differ by more than a certain amount (the same tolerance is used for all industries within a year, but has varied over time), then the Census compares each of them to annual payroll for the same plant and then takes the "best" one. The "best" one is selected in a form similar to the regression imputes, described below.

Administrative edits are similar to logical edits, but differ in that the alternative source of information is from administrative records. For instance, for payroll the administrative records come from IRS payroll tax records. Again, if the administrative data differ from the reported values, the reported values may be replaced.

Regression imputes are used to edit data when alternative sources of information are not available. The U.S. Census Bureau uses a variety of industry-specific regression-based imputation strategies. Since they do not require any observed alternative value, regression imputes are also used to impute missing values when no other information is available for a given variable. In general, regression imputes create predictions using one other variable, and (for plants surveyed in the Annual Survey of Manufactures), one-year lags of the imputed variable as well. Unlike administrative and logical edits, there is not necessarily any direct evidence that the value reported by the the establishment may be

incorrect.[25]

In order to measure the importance of each edit, we undertake the following exercise. For each type of edit, we replace *all* of an establishment's information with the clean data if it was affected by the edit. For example, to understand the importance of logical edits for total value of shipments, we use the cleaned outcomes for plants whose total value of shipments has a logical edit flag, and the raw outcomes for the other plants. We show how much measured misallocation in the U.S. is affected by each edit by showing its Shapley value.[26] For this context, the Shapley value is the value of the following thought exercise: we first consider all possible combinations of edits, done one at a time. We credit each edit for its marginal contribution within each (ordered) combination, and calculate the Shapley Value as the average of those marginal contributions. The results are in Table **??**, which we summarize graphically in Figures 6-**??**. The mapping from how we categorize the edit to the actual edit flag are in Appendix Table 7. We define "replicable" edits as ones which, generously, could in principle be done in the Indian data.

For misallocation, analyst corrections and logical imputes for shipments are each responsible for around a fifth of the gain in measured allocative efficiency, with logical and regression imputes for materials each responsible for around another sixth. Some categories matter for one particular year, for instance miscellaneous edits are responsible for a fifth of the measured gains in 2002, but aren't as relevant in 2007 and 2012. The rankings for the distributions of TFPQ and TFPR are similar. The importance of the analyst corrections are increasing over time; in 2012 they are responsible for the majority of the decline in dispersion for TFPR and TFPQ. Imputes for missing data barely matter for measured misallocation, and are among the least important categories for the distribution of TFPQ

[25] There may be strong *indirect* evidence that a reported value is incorrect, such as, hypothetically, if a plant reported production worker wages of $5 billion and only 1 production worker.

[26] For disclosure reasons, we report values for no trimming (for Gross Output). This is because with trimming, the exact sample varies depending on the edits used, leading to complicated implicit samples. The qualitative rankings of the importance of each edit is similar across trimming types.

and TFPR.

The results of Table 5 are unsatisfying - cross-country comparisons of measured misallocation in datasets which have been cleaned differently will pick up differences due to both underlying cross-country differences and cross-country differences in data cleaning. However, while comparing raw data solves the latter problem, it does so at the expense of introducing new errors. The natural solution is to compare datasets which have been commonly cleaned. While one strategy may be to use the approach of the U.S. Census Bureau everywhere, Table **??** shows that around $^2/_3$ of the changes - the logical imputes, analyst corrections, and administrative record edits - are difficult if not impossible to replicate in other contexts (depending on the availability of alternative reports for the same outcome and industry specialists). As an alternative, in section V we describe and then implement an algorithm for editing and imputing raw establishment-level information.

V An approach to Cleaning Firm-level Data

Establishment i reports p characteristics, $\boldsymbol{y}_i = \{y_{i1}, y_{i2} \dots y_{ip}\}$ (where items could be missing). The corresponding true values are $\boldsymbol{x}_i = \{x_{i1}, x_{i2} \dots x_{ip}\}$, with s_{ij} indicating if response j for establishment i is incorrect. Given the dataset of reported values $\boldsymbol{Y} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \dots \boldsymbol{y}_n\}$ the goal of data cleaning is to replace the faulty values so that (i) the cleaned data for each record i is plausible and internally consistent and (ii) the cleaned dataset is drawn from the same joint distribution as the true values $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \dots \boldsymbol{x}_n\}$. In order to do this, we use the approach of Kim et al. (2015), which we now describe.

First, we define the feasible region $\mathcal{D}$ of plausible reports. This region limits possible values by two types of rules: balance rules which require entries to add up[27] and a set of ratio edit rules which bound the ratios of any two variables. While the balance rules are *a*

[27] For instance, non-production wages + production wages = total wages, or more generally $\left(x_{iT_\ell} - \sum_{j \in \beta_\ell} x_{ij} = 0\right)$ for x_{iT_ℓ} as the total for the ℓth balance rule for the set of component variables β_ℓ.

priori clear, the ratio edit rules can come either from industry specific knowledge, or from outliers in the data itself. Fellegi and Holt (1976) note that the set of explicit ratio edit rules can imply additional ones as well.[28] While $\boldsymbol{s_i}$ is not directly observed, A_i indexes the failed ratio & balance edit rules. If, e.g., y_{i1} fails multiple edits and y_{i2} fails only one, then, other things equal, y_{i1} is more likely to be faulty than y_{i2}.

After cleaning the data, we want our cleaned data to be likely given a model for reporting error, likely given a model for error indicators, and likely given a model for the underlying data. More formally,

$$f\left(\boldsymbol{x_i}, \boldsymbol{s_i} | \boldsymbol{y_i}, A_i\right) \propto f\left(\boldsymbol{y_i} | \boldsymbol{x_i}, \boldsymbol{s_i}, A_i\right) f\left(\boldsymbol{s_i}, A_i | \boldsymbol{x_i}\right) f\left(\boldsymbol{x_i}\right). \tag{6}$$

For the model of reporting error, we maintain the U.S. Census Bureau's (implicit) approach to ratio and balance edits: data reported with error provides no information on the true value.[29] Therefore, $f\left(\boldsymbol{y_i} | \boldsymbol{x_i}, \boldsymbol{s_i}, A_i\right)$ is uniform over the support of feasible values if $y_{ij} \neq x_{ij}$.

However, unlike the Census Bureau, we also assume a uniform distribution for the errors. That is to say, we do not use weights on which variables are more likely to be reported with error, so all candidates $\boldsymbol{s_i}$ that result in feasible solutions are equally likely.

For the model for the underlying data, we assume that each establishment belongs to one of K mixture components (z). After assuming K,[30] we need to estimate the probability of membership in each component (π), and within each component the mean vector $(\boldsymbol{\mu})$ and covariance matrix $(\boldsymbol{\Sigma})$. In order to ensure that all of the draws will pass both the balance and ratio edits, we impose that the distribution of $\boldsymbol{x_i}$ conditional on μ, Σ, z_i, given

[28] For instance, rules $x_1 \leq x_2$ and $x_2 \leq x_3$ imply $x_1 \leq x_3$.

[29] One important exception to this rule is units errors (a.k.a. "rounded" edits), where the original reported value is divided by 1000 and then rounded to the nearest unit. Our approach did poorly replicating those edits (partially because we imposed a flat prior for which variable was more likely to be reported with error), and so we accepted all of the rounding edits and otherwise used raw data.

[30] In practice we set K=50, which is large enough that no data are in the lowest probability components.

feasible region $\mathcal{D}$ is

$$f\left(\boldsymbol{x}_i|\theta_i\right) = \mathcal{N}\left(\boldsymbol{x}_{i,NT}|\boldsymbol{\mu}_{z_i}, \boldsymbol{\Sigma}_{z_i}\right) \prod_{l=1}^{q} \delta\left(x_{iT_\ell} - \sum_{j \in \beta_\ell} x_{ij}\right) \mathbb{1}\left[\boldsymbol{x}_i \in \mathcal{D}\right]$$

where $\delta\left(\cdot\right)$ is that Dirac delta function with the point mass at zero and $\boldsymbol{x}_{i,NT}$ is the set of reported values which are themselves totals of other reported values.

For each 6-digit NAICS industry-year, we run a single chain of Markov Chain Monte Carlo with a burn-in of 2000 iterations, and then 5000 additional iterations, keep the data from each 1000th iteration (for a total of 5 completed datasets for each industry-year in each country). Each iteration consists of first proposing $\boldsymbol{s_i}$ which are consistent with $\boldsymbol{A_i}$, and then editing values $\boldsymbol{y_i}$ given the draw of $\boldsymbol{s_i}$ and the underlying probability distributions for the responses which were not reported with error.

VI Cross Country Differences in Measured Misallocation

The main choice associated with the model in Section V is defining the feasible region $\mathcal{D}$. For the U.S. data, we use the industry-year-specific ratio edit rules (upper and lower bounds) used by the Census Bureau for the 2002, 2007, and 2012 Censuses of Manufacturers.

For the Indian data, we define the feasible region by following the resistant fences method, which is the starting point for how Census chooses its ratio bounds (Thompson and Sigman, 1999). Within each industry, for each log ratio $r_{jk} = \ln\left(\frac{y_j}{y_k}\right)$, we calculate its 25th and 75th percentiles, Q_{jk}^{25} and Q_{jk}^{75}, and therefore the interquartile range IQR_{jk} . We then flag all ratios that are either smaller than $Q_{jk}^{25} - C \times IQR_{jk}$ or larger than $Q_{jk}^{75} + C \times IQR_{jk}$ where C is a pre-specified threshold. The variables we use are the total cost of materials, total value of shipments, number of blue and white collar workers, blue and white collar wages and (in India) the sampling weight, and we run the estimation separately by industry (6 digit in the U.S. and 2 digit in India) and year. We calculate

allocative efficiency separately for each year. In Table 6, we report the average of the 2000-2011 results for India with $C = 3$ and for 2002, 2007, and 2012 in the US. For comparison, we also report results from the Captured Data with 1% and 2% trimming.

The gap between the raw U.S. and raw Indian data shrinks after applying the common data editing procedure. In the commonly-cleaned data with 1% tail trimming, allocative efficiency in the U.S. in 2002 is slightly lower than the Indian time series average (0.499 vs 0.521). In 2007 and 2012, allocative efficiency in the US is, respectively, 0.161 and 0.231—significantly lower than the Indian average. With 2% trimming, allocative efficiency is slightly higher in 2002 in the US vs India, and about 30% lower in 2007 and 2012.

VII Discussion

In this paper, we use previously unexplored versions of the United States Census of Manufactures for 2002, 2007, and 2012 in order to investigate the role that measurement plays for estimating misallocation. We have two complementary goals. The first is to quantify the importance of data cleaning. We show that in the data that is reported directly to the Census Bureau by U.S. establishments, measured TFP in the United States manufacturing sector is less than an eighth of what it would be if it had the same level of allocative efficiency as the Indian manufacturing sector. We do not take this result literally: there are many reasons to believe that comparing the raw U.S. data to its counterparts in India or other countries is not like-for-like.[31] We also see large differences for other measures of dispersion, such as the interquartile range. While editing matters more for younger and smaller firms, even the oldest and largest ones have their responses relatively heavily edited.

[31] For no country do we know if measured misallocation in the original establishment responses is larger or smaller than it is in reality, nor do we have a way of comparing the relative precision of self-reported information across countries. We do know that the vast majority of Indian firms are unable to fill out their survey forms on a computer.

Many of the important edits undertaken in the U.S. are infeasible for researchers using other datasets, because they either use multiple responses for the same information or because they rely on U.S. Census Bureau industry experts. When we use common data cleaning strategies, we find little or no evidence that allocative efficiency is significantly lower in India than in the United States, and for 2007 and 2012 we find that allocative efficiency is significantly *higher* in India than in the US.

One important limitation of our approach highlighting the effect of data processing done by the U.S. Census Bureau is that it does not provide any insight into why there appears to be so much measurement error in the United States (nor do we have reduced form evidence that there is more (or less) measurement error in the U.S. relative to other countries).

Nevertheless, there is a large scope for different measurement choices to affect the estimation of misallocation and dispersion in manufacturing, and we show that the results are sensitive to those choices. Neither ad-hoc approaches nor blissful ignorance are necessary or desirable. In this project, and in ongoing work, we show the value in working with experts in data cleaning (who currently tend to associate with other disciplines). To that end, we suggest an alternative approach for cleaning establishment-level data using a hierarchical Bayesian approach. In addition to being a consistent approach for comparing cross-country differences in outcomes, the method is more broadly useful for data cleaning instead of more traditional ad-hoc approaches such as winsorizing.

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A Appendix

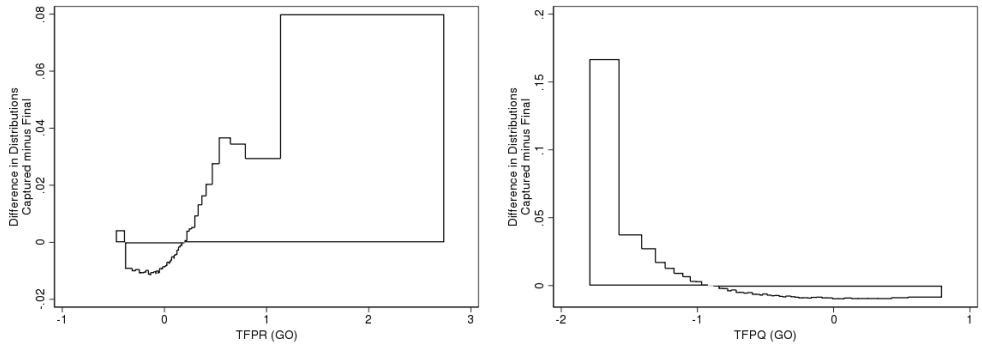
A.I Cross-Country Estimates of Misallocation

For India, we use the Annual Survey of Industries (the ASI). Factories with over 100 workers are surveyed every year, while smaller establishments are surveyed every few years (the ASI is designed to be representative at the State by Industry level, so establishments without local competitors are more likely to be surveyed). Hsieh and Klenow (2009) use

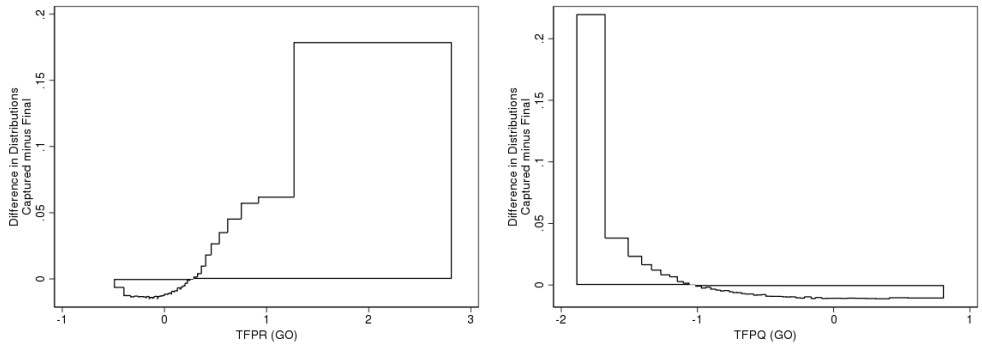
the same dataset, and we follow standard practice in generating measures of gross output, intermediate inputs, capital, and payroll (Allcott et al., 2016). Industries are grouped using India's NIC (National Industrial Classification) codes, and we report the value of reallocation for 2009. For the U.S. and India, we use cost-shares from the NBER-CES Manufacturing Industry Database as our measures of industry production elasticities, and multiply the book value of capital by 10% in order to impute the cost of capital.

Figure 1: Difference in Densities between Raw and Cleaned Data

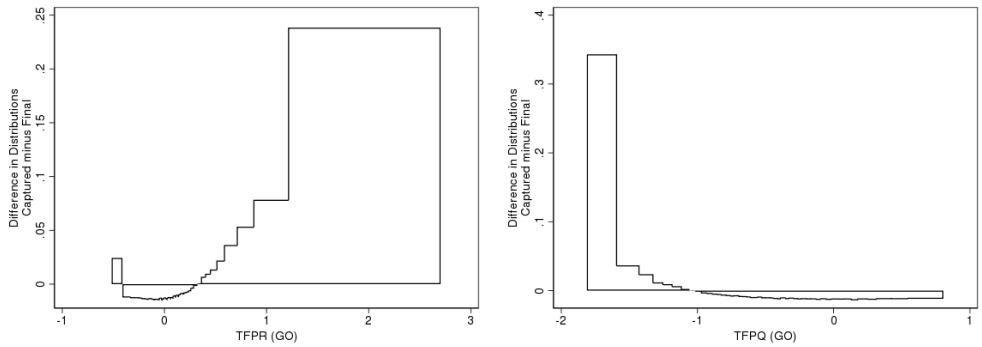
Panel A: 2002



Panel B: 2007



Panel C: 2012

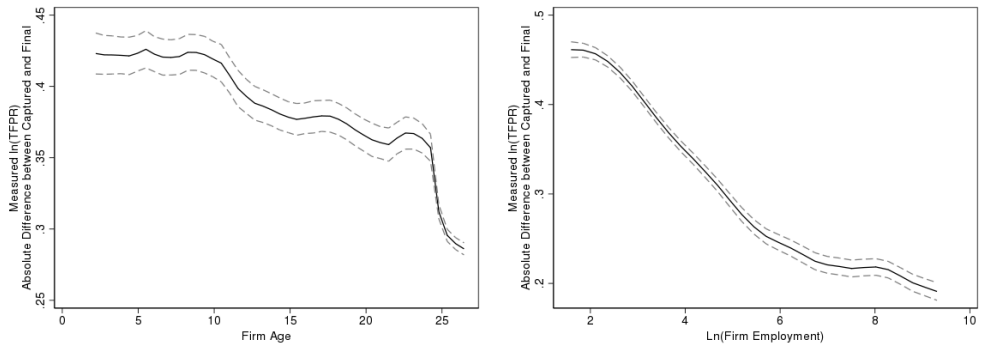


Source: U.S. Census of Manufacturers.

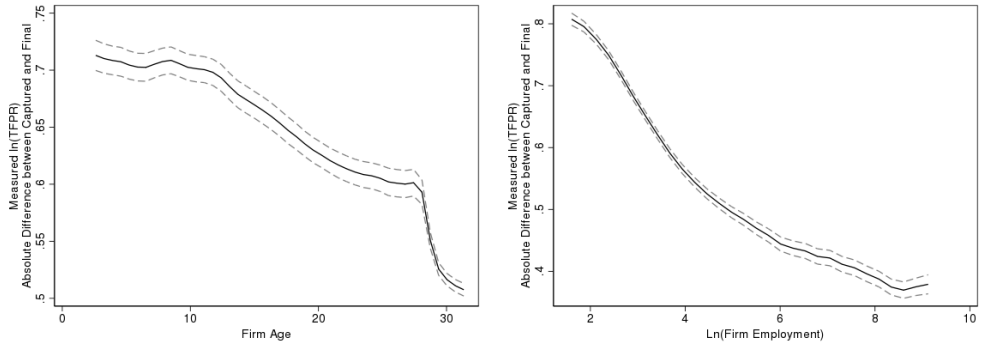
Notes: Each figure shows the difference in densities between the raw and cleaned data in the U.S. Census of Manufacturers, with each panel representing a different year, and the left figures showing (log, industry-scaled) TFPR and the right figures showing TFPQ. The cleaned data is binned into 50 groups, each representing 2% of the density of the distribution. The end-points of each bin reflect the high and low values for the plant characteristic ($\pm$10% for disclosure avoidance). The y axis reflects the difference in densities for the raw data relative to the clean data, with positive values for when there are more plants in that bin in the raw data. The sum over all of the bins is therefore 0.

Figure 2: Absolute Relationship between firm size/age & editing (TFPR)

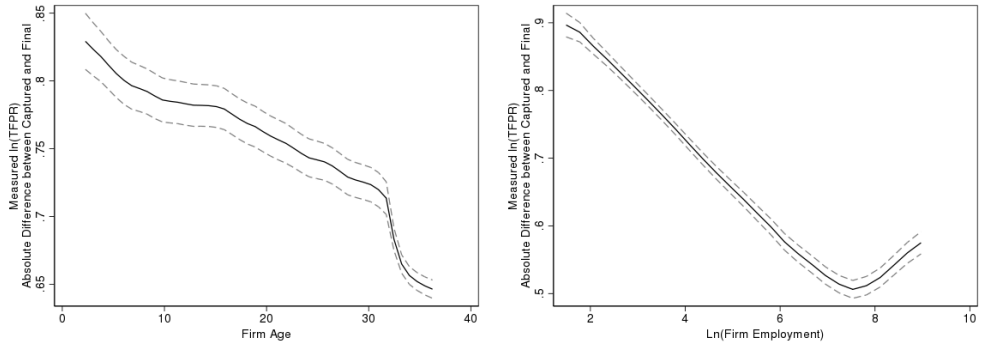
Panel A: 2002



Panel B: 2007



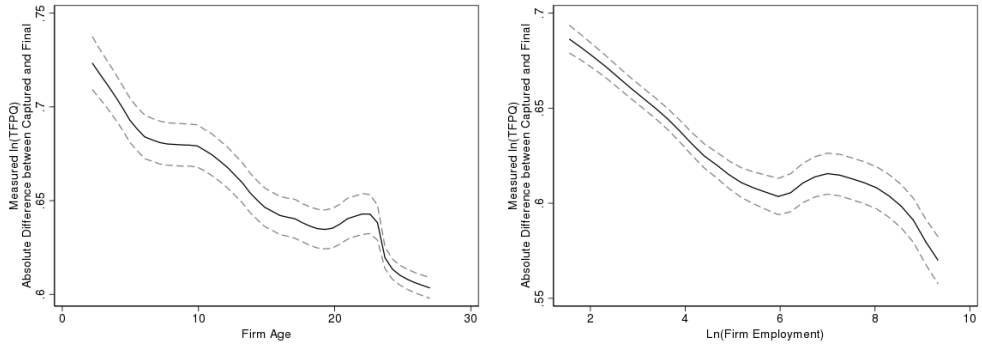
Panel C: 2012



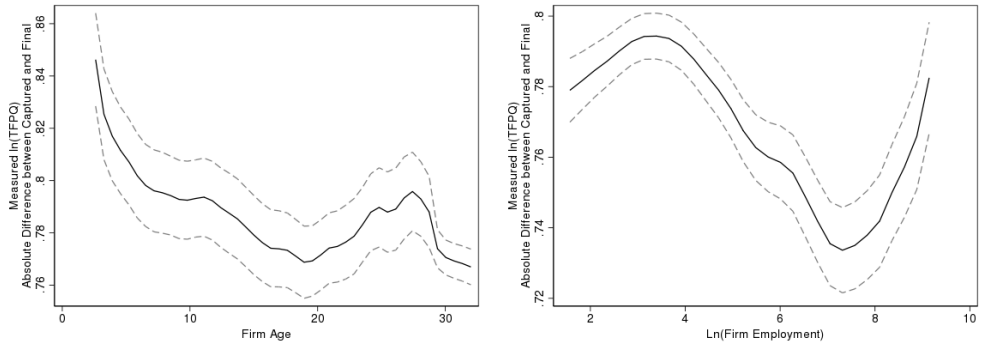
Source: U.S. Census of Manufacturers.
Notes: Each figure shows the relationship between size and the extent of data cleaning of TFPR in the U.S. Census of Manufacturers, with each panel representing a different year, and the left figures showing firm age and the right figures showing firm employment. All figures plot smoothed values from kernel-weighted local polynomial regressions of the absolute difference between raw and clean TFPR, and firm size. We use the Epanechnikov kernel with the Stata default bandwidth.

Figure 3: Absolute Relationship between firm size/age & editing (TFPQ)

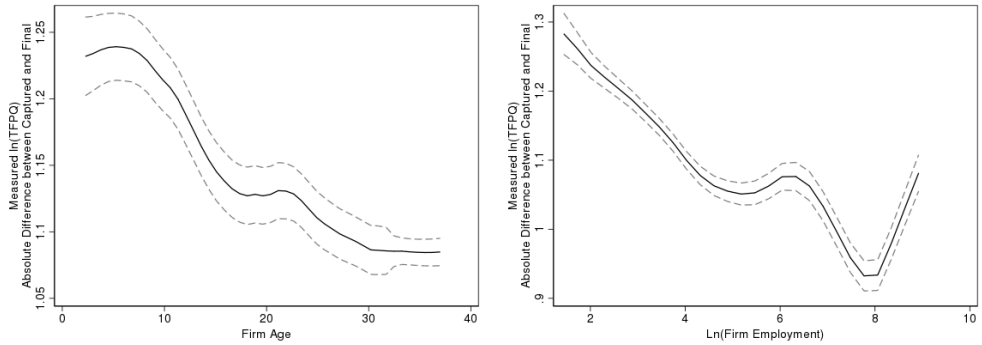
Panel A: 2002



Panel B: 2007



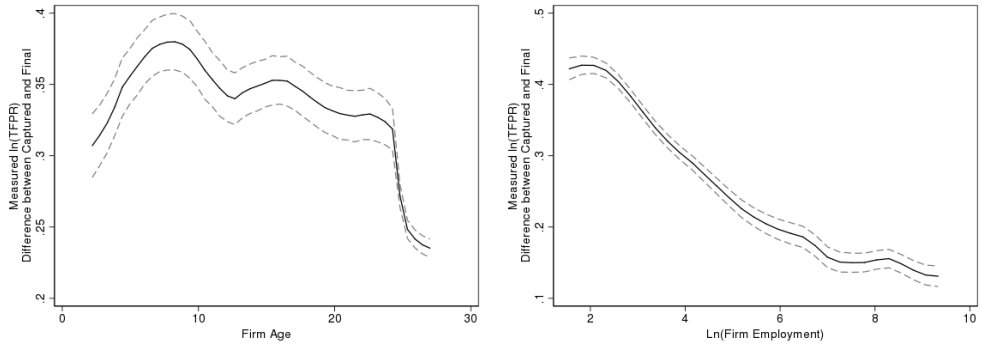
Panel C: 2012



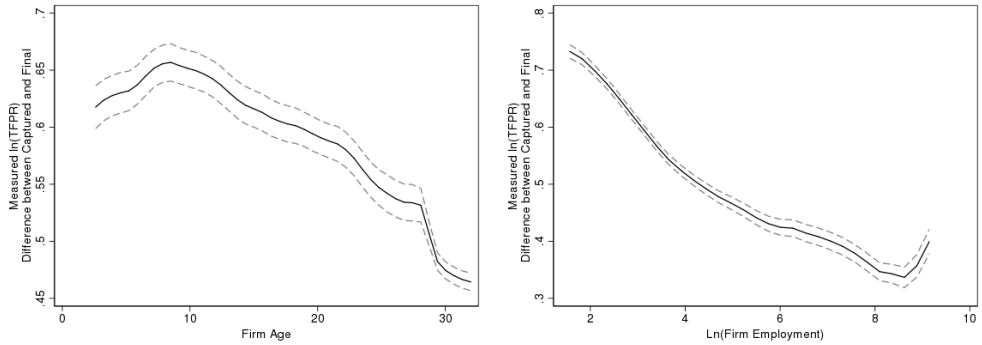
Source: U.S. Census of Manufacturers.
Notes: Each figure shows the relationship between size and the extent of data cleaning of TFPQ in the U.S. Census of Manufacturers, with each panel representing a different year, and the left figures showing firm age and the right figures showing firm employment. All figures plot smoothed values from kernel-weighted local polynomial regressions of the absolute difference between raw and clean TFPQ, and firm size. We use the Epanechnikov kernel with the Stata default bandwidth.

Figure 4: Relationship between firm size/age & editing (TFPR)

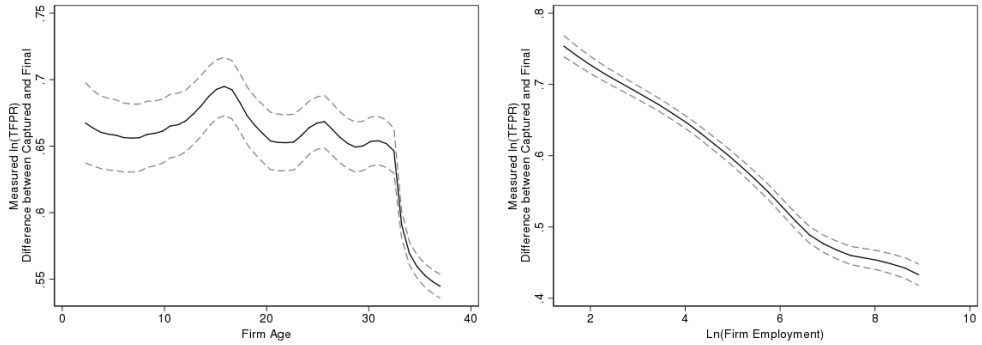
Panel A: 2002



Panel B: 2007



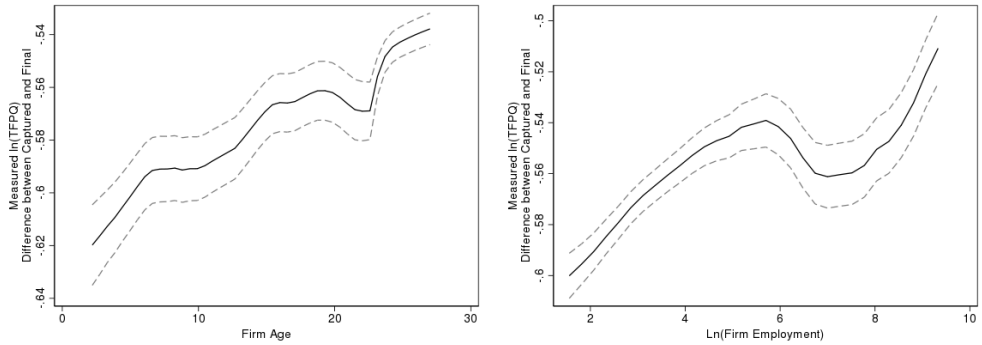
Panel C: 2012



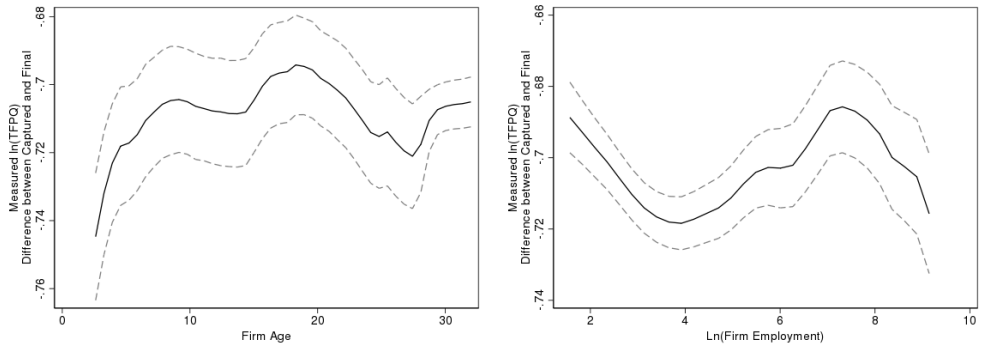
Source: U.S. Census of Manufacturers.
Notes: Each figure shows the relationship between size and the extent of data cleaning of TFPR in the U.S. Census of Manufacturers, with each panel representing a different year, and the left figures showing firm age and the right figures showing firm employment. All figures plot smoothed values from kernel-weighted local polynomial regressions of the difference between raw and clean TFPR, and firm size. We use the Epanechnikov kernel with the Stata default bandwidth.

Figure 5: Relationship between firm size/age & editing (TFPQ)

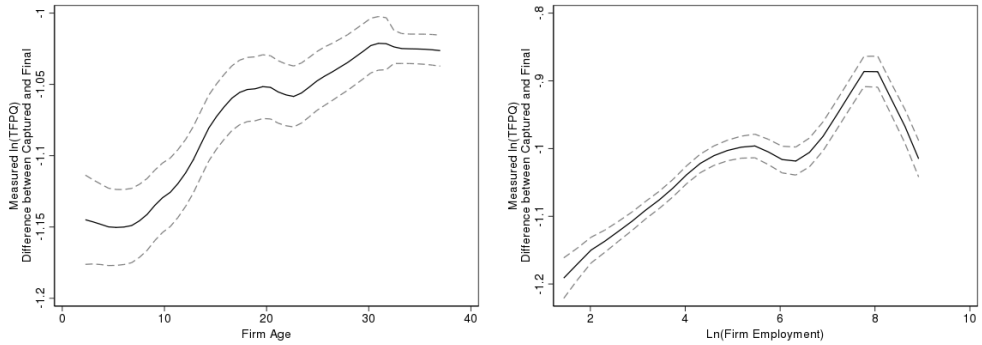
Panel A: 2002



Panel B: 2007

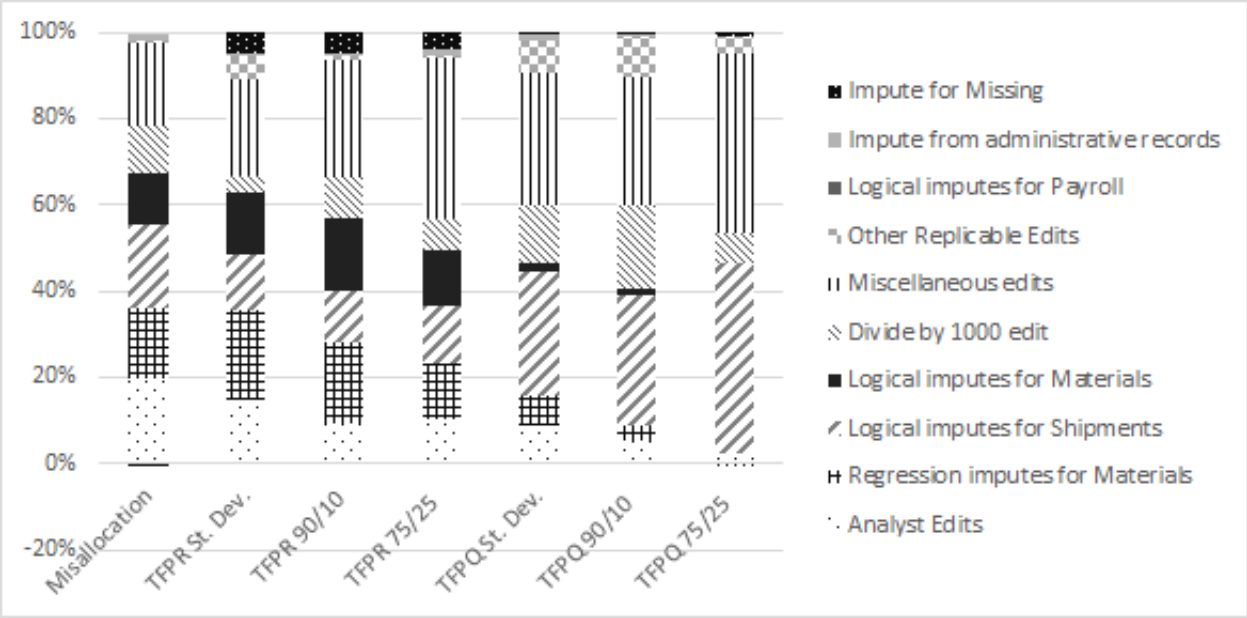


Panel C: 2012



Source: U.S. Census of Manufacturers.
Notes: Each figure shows the relationship between size and the extent of data cleaning of TFPQ in the U.S. Census of Manufacturers, with each panel representing a different year, and the left figures showing firm age and the right figures showing firm employment. All figures plot smoothed values from kernel-weighted local polynomial regressions of the difference between raw and clean TFPQ, and firm size. We use the Epanechnikov kernel with the Stata default bandwidth.

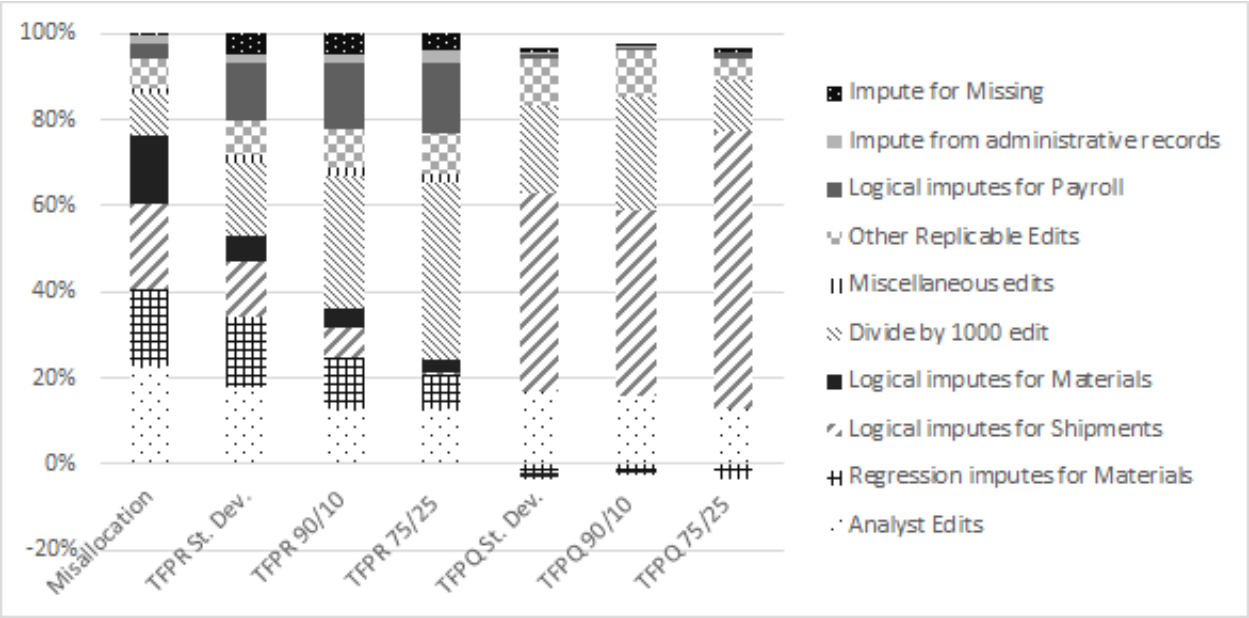
Figure 6: Edits' Shapley Share of Total Effect, 2002



Source: U.S. Census of Manufactures

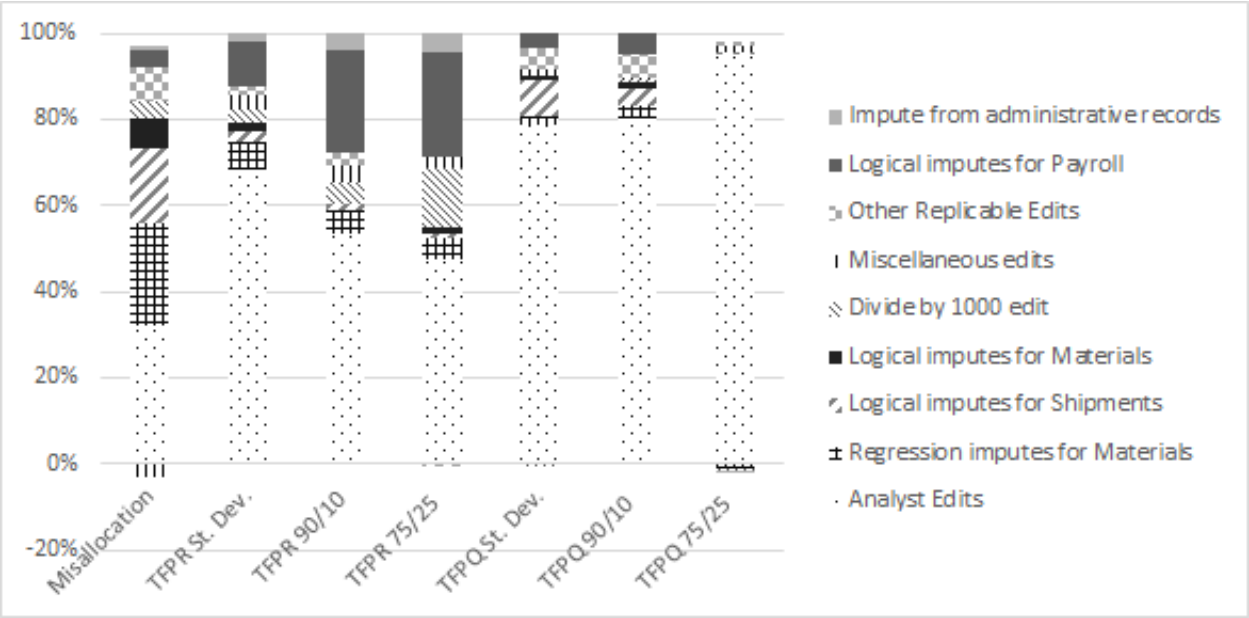
Notes: Our model for calculating allocative efficiency is as described in the text. For each year, we calculate allocative efficiency using all of the possible permutations of including some edits. We then report the average marginal share of the total change, following Shapley (1953). The absolute changes are in Table 3. We report changes for 0% trimming.

Figure 7: Edits' Shapley Share of Total Effect, 2007



Source: U.S. Census of Manufacturers.
Notes: See notes for Figure 6

Figure 8: Edits' Shapley Share of Total Effect, 2012



Source: U.S. Census of Manufacturers.
Notes: See notes for Figure 6

Table 1: Edits Made to the U.S. Census of Manufacturers

Edit/Impute Action	Occurs when...
Administrative (A)	The item is imputed by direct substitution of corresponding administrative data (for the same establishment/record).
Cold Deck Statistical (B)	The item is imputed from a statistical (regression/beta) model based on historic data.
Analyst Corrected (C)	The reported value fails an edit, and an analyst directly corrects the value.
Model (Donor) Record (D)	The item is imputed using hot deck methods.
Receipts (F)	Receipts posted from the nonemployer universe as a result of the Business Master File EIN/SSN cross-reference match or primary SSN (one-for-one match).
High/Low (E)	The item is imputed with a value near the endpoints of the imputation range.
Historic (H)	The item is imputed by using historic ratio data for the same establishment.
Subject Matter Rule (J)	The item is imputed using a subject matter defined rule (e.g. y=1/2x).
Prior Year Ratio (HQ)	The item is imputed using a ratio of historic data times current reported values.
Raked (K)	The sum of a set of detail items do not balance to the total. The details are then changed proportionally to correct the imbalance.
Logical (L)	The item's imputation value is defined by an additive mathematical relationship (e.g., obtaining a missing detail item by subtraction).
Midpoint (M)	The item is imputed by direct substitution of midpoint of imputation range.
Rounded (N)	The reported value is replaced by its original value divided by 1000.
Prior Year Administrative (P)	The item is imputed by ratio imputation using corresponding administrative data from prior year (for same establishment).
Direct Substitution (S)	The item is imputed by direct substitution of another item's value (from within the same questionnaire.)
Trim-and-Adjusted (T)	The item was imputed using the Trim-and Adjust balancing algorithm.
Unable to Impute (U)	The reported item is blank or fails an edit, and the system cannot successfully substitute a statistically reasonable value for the original data.
Industry Average (V)	The item is imputed by ratio imputation using an industry average.
Warm Deck Statistical (W)	The item is imputed from a statistical (regression/beta) model based on current data.
Unusable (X)	The sum of a set of detail items cannot be balanced to the total because none of the scripted solutions achieved a balance.
Data Impute (2)	Data item imputed from a reported data item
Payroll Quarterly (4)	First quarter payroll reported higher than annual payroll; data adjusted during general data prep or legacy edits

Notes: Edit/imputation descriptions for the U.S. Census of Manufacturers from Grim (2011) and White et al. (2018), with a few additional categories.

Table 2: Dispersion of TFPR and TFPQ in the U.S. Census of Manufactures

	Captured Data			Census-Cleaned Data		
	Outcome			Outcome		
Year	St. Dev	90/10	75/25	St. Dev	90/10	75/25
Panel A: TFPR						
2002	0.889	1.337	0.577	0.401	0.783	0.331
2007	0.955	1.716	0.902	0.442	0.87	0.356
2012	1.089	1.888	1.031	0.421	0.831	0.346
Panel B: TFPQ						
2002	3.059	7.266	5.705	0.761	1.828	0.924
2007	2.669	6.602	4.761	0.947	2.337	1.078
2012	3.248	7.44	5.205	0.741	1.8	0.896

Source: U.S. Census of Manufactures
Notes: TFPR and TFPQ are calculated as described in the text. For each of the two underlying data sources (firm responses with no edits, and final-cleaned data), we report the Standard Deviation, the 90/10 and 75/25 ratio for each outcome.

Table 3: Measured Allocative Efficiency in the U.S. Census of Manufactures

	Captured Data			Census-Cleaned Data		
	Trimming %			Trimming %		
Year	0%	1%	2%	0%	1%	2%
Panel A: Gross Output						
2002	0.00005	0.109	0.176	0.14	0.461	0.554
2007	0.000005	0.012	0.024	0.042	0.302	0.425
2012	0.00000038	0.004	0.024	0.059	0.349	0.455
Panel B: Value Added						
2002	0.021	0.49	0.55	0.452	0.649	0.714
2007	0.026	0.316	0.435	0.395	0.62	0.673
2012	0.005	0.274	0.392	0.452	0.652	0.696

Source: U.S. Census of Manufactures
Notes: Our model for calculating Allocative Efficiency is as described in the text. For each of the two underlying data sources (firm responses with no edits, and final-cleaned data), we report measured allocative efficiency for different values of trimming extreme values of (industry-scaled) TFPQ and TFPR.

Table 4: Measured Allocative Efficiency in the U.S. Annual Survey of Manufactures

	Captured Data			Census-Cleaned Data		
	Trimming %			Trimming %		
Year	0%	1%	2%	0%	1%	2%
Panel A: Gross Output						
2002	0.003	0.209	0.415	0.160	0.458	0.555
2007	0.00004	0.026	0.058	0.085	0.294	0.416
2012	0.00007	0.004	0.074	0.077	0.340	0.457
Panel B: Value Added						
2002	0.061	0.470	0.624	0.469	0.620	0.669
2007	0.101	0.442	0.503	0.431	0.585	0.637
2012	0.071	0.396	D	0.461	0.623	0.666

Source: U.S. Census of Manufactures, using only plants in the Annual Survey of Manufactures sample
Notes: Our model for calculating Allocative Efficiency is as described in the text. For each of the two underlying data sources (firm responses with no edits, and final-cleaned data), we report measured allocative efficiency for different values of trimming extreme values of (industry-scaled) TFPQ and TFPR. One value is marked "D" because, due to small differences between samples, we could not disclose the 2012 value added number with 2% trimming.

Table 5: India / US Comparison of Measured Allocative Efficiency

	Allocative Efficiency	Indian AE relative to 2002-2012 Average in:	
Year	India	Census-cleaned US	Captured US
2002	0.329	1.12	12.65
2007	0.323	.95	80.75

Source: U.S. Census of Manufactures and Indian Annual Survey of Industry
Notes: Our model for calculating allocative efficiency is as described in the text. For each of the years for which we have overlapping data (2002 and 2007), we report measured allocative efficiency in India, as well as the ratio of measured Allocative Efficiency in India divided by the average value in the U.S., where numbers larger than one imply higher allocative efficiency in India. Values trim 1% extremes of TFPQ and TFPR.

Table 6: Edits' Shapley Share of Total Effect

			TFPR			TFPQ		
Edit	Year	Misallocation	St. Dev	90/10	75/25	St. Dev	90/10	75/25
Analyst Edits	2002	0.197	0.145	0.094	0.104	0.086	0.055	0.026
Analyst Edits	2007	0.230	0.178	0.128	0.129	0.181	0.169	0.135
Analyst Edits	2012	0.337	0.663	0.517	0.462	0.784	0.800	0.982
Regression imputes for Materials	2002	0.166	0.211	0.185	0.129	0.073	0.034	-0.002
Regression imputes for Materials	2007	0.175	0.161	0.119	0.079	-0.023	-0.015	-0.031
Regression imputes for Materials	2012	0.252	0.067	0.053	0.048	0.020	0.027	-0.011
Logical imputes for Shipments	2002	0.195	0.128	0.120	0.135	0.285	0.302	0.439
Logical imputes for Shipments	2007	0.199	0.133	0.068	0.007	0.491	0.451	0.692
Logical imputes for Shipments	2012	0.183	0.024	0.011	0.010	0.083	0.040	0.003
Logical imputes for Materials	2002	0.124	0.142	0.171	0.128	0.020	0.014	0.004
Logical imputes for Materials	2007	0.162	0.056	0.048	0.031	-0.011	-0.005	0.000
Logical imputes for Materials	2012	0.074	0.017	0.001	0.017	0.009	0.017	0.001
Divide by 1000 edit	2002	0.109	0.044	0.092	0.075	0.134	0.192	0.070
Divide by 1000 edit	2007	0.098	0.170	0.305	0.410	0.217	0.275	0.128
Divide by 1000 edit	2012	0.044	0.027	0.048	0.135	0.000	0.002	-0.006
Miscellaneous edits	2002	0.195	0.220	0.275	0.369	0.308	0.298	0.417
Miscellaneous edits	2007	0.009	0.019	0.020	0.019	-0.001	-0.003	-0.004
Miscellaneous edits	2012	-0.032	0.035	0.038	0.025	0.013	0.008	0.015
Other Replicable Edits	2002	0.002	0.054	0.009	0.004	0.077	0.095	0.038
Other Replicable Edits	2007	0.069	0.083	0.091	0.094	0.119	0.111	0.054
Other Replicable Edits	2012	0.082	0.018	0.028	0.000	0.052	0.054	0.011
Logical imputes for Payroll	2002							
Logical imputes for Payroll	2007	0.035	0.132	0.152	0.164	0.011	0.007	0.012
Logical imputes for Payroll	2012	0.043	0.102	0.232	0.238	0.031	0.047	-0.001
Impute from administrative records	2002	0.017	0.004	0.003	0.015	0.012	0.008	0.002
Impute from administrative records	2007	0.020	0.017	0.020	0.029	0.004	0.004	0.004
Impute from administrative records	2012	0.008	0.020	0.036	0.041	0.001	0.002	0.000
Impute for Missing	2002	-0.004	0.051	0.051	0.041	0.006	0.003	0.006
Impute for Missing	2007	0.003	0.050	0.049	0.040	0.012	0.007	0.009
Impute for Missing	2012	0.008	0.027	0.035	0.025	0.007	0.004	0.005

Source: U.S. Census of Manufactures
Notes: Our model for calculating allocative efficiency is as described in the text. For each year, we calculate allocative efficiency using all of the possible permutations of including some edits. We then report the average marginal share of the total change, following Shapley (1953). The absolute changes are in Table 3, we report changes for 0% trimming.

Table 7: Measured Allocative Efficiency After Bayesian Editing and Imputation

		Trimming %		
Country	Year(s)	0%	1%	2%
U.S.	2002	0.257	0.5	0.582
U.S	2007	0.037	0.161	0.369
U.S.	2012	0.029	0.231	0.385
India	2000-2011	0.465	0.521	0.554

Source: U.S. Census of Manufactures and Indian Annual Survey of Industry
Notes: Our model for calculating allocative efficiency is as described in the text. For each year, we identify values to edit and how to replace them with imputations following the method described in the text. In each column we then (potentially) further trim outliers.

Table 8: Categorization of Edits Made to the U.S. Census of Manufacturers

Edit/Impute Label	For edit flag
Analyst Edits	Any variable in (C)
Impute from administrative records	Any variable in (A)
Divide by 1000 edit	Any variable in (N)
Impute for Missing	Any flag with no originally reported value
Logical imputes for Materials	Materials in (L)
Payroll logical impute	Payroll in (L)
Regression imputes for Materials	Materials in (B) or (W)
Shipments logical impute	Shipments in (L)
Other Replicable Edits	"Shipments in {(B), (E), (G), (H), (J), (M), (S), (V), (X)} or Materials in {(E), (H), (M)} or Payroll in {(H), (HQ), (J), (G), (2), (4)}
Not Elsewhere Classified	The remainder

Notes: Edit rules are defined in Table 1. Categorization done by authors.